What a Missing-Relation Cell Requires: Four Candidate Relations and a Negative Result Under Physical Contact

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Draft note, not for submission. The bibliography has not been verified against published records. No confirmatory run was performed: all physical evidence is calibration, and the preregistration excludes calibration from confirmatory estimates. Section 7 states precisely which claims this record does and does not support. Remove this box by setting `\draftnotefalse`.

Abstract

Prior work in this program showed that structured failure after parameter repair can justify a prepared increase in target-model *order*—a missing dynamic mode. The natural next cell in the taxonomy is a missing *relation*: a residual conditioned not on an unrepresented state variable of the target, but on a second entity. We preregistered a design for that cell, built it, and report a negative result.

We evaluated four candidate relations against three requirements fixed in advance: the relation's effect must exceed the task tolerance, a single-entity model of the target must not suffice, and the relation must not be reducible to a dynamic mode. **Collision** fails the first—a struck target's displacement per contact is of the order of the interaction radius, five independent measurements placing it below the 20 mm placement criterion. **Carriage** fails the second—a model observing only the target's own trajectory matched the relational arm exactly. **Capture**—a body arrives at a still target and carries it off—satisfies all three under injected coupling, where the relational arm scores 0.650 against a single-entity competitor's 0.000 across 200 preregistered paired cells (one-sided exact McNemar, 130 discordant pairs, none the competitor's, $p = 7.3 \times 10^{-40}$).

Under **physical contact** in Isaac Sim it fails the third. Across 107 cells and three configurations of a surgical needle driver, the constant-velocity mode operator lands **0.957 to 1.000** of cells while the relational arm lands 0.174 to 0.583 and **never wins a paired cell that the single-entity arm loses** (0 against 4, twice). The relation is made necessary by *intermittency*, and the arm requires a median of 22 steps to come to rest after its goal stops, against a commanded pause of 4 and 54 steps of grip on the object: the pause never begins.

The contribution is the requirement set, the measurement that each candidate fails, and a physical criterion recovered by sweeping the carrier's settling time under injected coupling: the mode operator returns once settling reaches the carrier's duty-cycle period, 0.133 to 0.917 across that threshold. The same sweep refutes the tidier account we started with—the relational arm does *not* collapse under a smoothed carrier, so settling explains the mode operator's return and not the relational arm's physical failure, which remains open.

Keywords: world models, structural adaptation, relational models, negative results, preregistration, embodied control, surgical simulation

1 Introduction

An embodied agent whose predictions fail can respond at three depths. It can repair parameters inside a fixed model class. It can expand the class to represent a state variable it lacked—*model-order expansion.*

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Or it can expand the class to represent a *dependence on another entity*, which is a change of a different kind: the model gains not a variable but an argument.

This paper concerns the third. The program's governing principle is that intelligence chooses the *smallest sufficient change*, so a relational expansion is warranted only where a mode expansion is insufficient. That is a demanding condition, and this paper's result is that it is more demanding than it appears.

Relational and object-centric dynamics models establish *how* such structure is represented [1, 7, 8]. This paper does not compete on that axis: it assumes a prepared relation-module operator and asks *when* such an operator becomes necessary. Classical model-order selection asks the analogous question one level down [9].

We report a negative result and a positive characterisation. The negative result is that the task family we built does not exhibit a missing relation under physical contact: the mode operator suffices. The positive characterisation is the requirement set that makes the negative result informative—three task-level requirements, each of which eliminated a candidate relation on measured grounds, and one physical requirement on the manipulator that emerged from the failure.

1.1 Why publish this

Three reasons, in ascending order of importance.

1. The requirement set is reusable. Anyone building a missing-relation benchmark must satisfy it, and three of the four requirements are cheap to check before any hardware is involved.
2. The mode/relation boundary is not where the taxonomy places it. A constant-velocity operator absorbed a coupling designed specifically to be outside it. That is a finding about the taxonomy, not about this scene.
3. The failure is a *shortcut*, not a null [4]. The relational task was solvable—the oracle arm lands 1.000 of physical cells—by a mechanism simpler than the one the benchmark names, and that mechanism was named in advance as the discriminating control.

2 Design, locked before measurement

The design was locked before any physical cell was run, and no locked rule was changed afterwards.

2.1 Task

A commitment point. The agent commits to an irreversible placement onto a target that a reference body captures and carries. The action takes `dispense_latency` steps and lands wherever the target is at completion; there is no correction. The structure is required, not incidental—a continuous reach-and-hold version produced no separation between any arms, because continuous re-aiming averages prediction error away.

Placement tolerance **20 mm**, on two grounds fixed before this study existed: the task family's own termination threshold, and the block's half-height.

2.2 Arms

Arm	Target model	Role
A	Frozen	Secondary baseline
B	Parameter repair, independent-entity model	Primary comparator
C	**Mode expansion—the prepared operator**	**Discriminating control**
SELF	**The target's own trajectory only**	**Decisive competitor**
D	Relation expansion	Primary intervention
D*	Relation with the true landing supplied	Diagnostic ceiling

All arms share controller, task, tolerance and commit policy, and differ only in the predicted landing point. Arm D falls back to arm B when its gate does not fire, so $D \geq B$ holds by construction; the informative quantities are how often D engages and how much it wins by when it does, and both are reported throughout.

SELF is deliberately **ungated**—arm D may act only when the relation-adequacy gate fires, SELF acts whenever its own pattern is identifiable. The asymmetry favours the competitor by design.

2.3 Requirements on the relation, fixed in advance

- **R1—the effect must exceed the tolerance.** The relation must move the target further than the task's own success criterion, or no arm can be distinguished by it.
- **R2—the relation must be necessary.** A model observing only the target's own trajectory must not suffice. This is what arm SELF tests.
- **R3—the relation must not reduce to a mode.** The second entity's influence must not be expressible as a dynamic mode of the target alone. This is what arm C tests.
- **R4—onset must be observable, or the claim must be scoped after onset.**

R4 was settled early and negatively, and the paper's claim is scoped accordingly: in the control worlds a body approaches the target and nothing happens, and it approaches *closer* than in the treatment—12 and 14 mm against 42 mm, in 1.00 of cells. Up to the moment of contact a capturing approach and a non-capturing one are the same observation, so no arm could predict onset and the paper does not claim to.

3 Four candidates, and where each fails

Relation	R1 effect > tol.	R2 necessary	R3 not a mode
Collision—struck and released	**no**	yes	yes
Carriage—rides throughout	yes	**no**	—
Capture, injected coupling	yes	yes	yes
Capture, physical contact	yes	**no**	**no**

3.1 Collision fails R1

The push moves the target away, which reduces penetration, which reduces the push. Displacement per contact therefore settles at the order of the interaction radius, below the 20 mm criterion. Measured five independent ways. This is a structural property of a penetration-driven contact law, not a tuning failure: raising the body's speed raises the equilibrium separation, not the displacement.

3.2 Carriage fails R2

A target that rides its reference throughout clears the tolerance easily and makes the relation unnecessary—a single-entity model of the target's own trajectory matched the relational arm **exactly**, cell for cell. The second entity supplies nothing the target's own history does not.

3.3 Capture satisfies all three under injected coupling

Capture—a body arrives at a *still* target and carries it off—was designed against both failures. Before the arrival the target is perfectly still, so its own history says nothing; afterwards it rides, so the effect accumulates without bound. It carries a third property that R2 turns on: the carrier moves **intermittently**, and the pauses are not predictable from the target alone.

R2, preregistered. The rule was fixed in writing before the SELF arm was implemented. On 200 paired cells at offsets [+4, +6], arm D scored **0.650 against SELF's 0.000**—130 discordant pairs, **none of them the competitor's**, one-sided exact McNemar $p = 7.3 \times 10^{-40}$. On the amended band [+4, +8] with fresh seeds, 0.735 against 0.045 and a discordant split of **146 to 8**.

All four arms on one run, which the two statements above deliberately are not—a preregistered pairwise comparison scores two arms, and reading a four-arm table off two runs with different seeds and different commit bands is how a figure in this project first went wrong:

CPU, injected, 60 cells	arm B	arm C	SELF	**arm D**
	0.000	0.133	0.083	**0.417**

R1 and R3 hold here. Arm B lands nothing, so the effect exceeds the tolerance; arm C lands 0.133, so the mode operator helps only partially, which is exactly the partial help the design requires and does not get under contact.

The competitor was not broken. It acted on 0.675 of cells, and its median miss *when acting* was 60.0 mm against 30.0 mm when it declined—it extrapolates through a pause it cannot see, which is exactly the information the relation supplies.

Two limits were measured with it, and both are reported rather than relied on. The single-entity arm catches up from commit offset **+30**, a little over two burst cycles after the arrival; the protocol's commit window is [−6, +6], where SELF scores ≤ 0.02. And what stops arm D first is its own gate rather than the competitor—the constant-velocity statistic climbs as the carry lengthens, and arm D declines exactly where it crosses the ceiling written to keep a drift control out.

3.4 Capture fails R2 and R3 under physical contact

The Isaac Sim lift scene and its needle variant, a surgical needle driver, with the same cell loop and the same arms. The simulator replaces the arithmetic: the cell commands the bodies, steps physics, and *reads* where the object went.

The scene produces the relation. **60 of 60 cells were captures** by the preregistered verdict, so the outcome the calibration pilot was written to catch—"the scene does not produce a capture at all"—did not occur.

Configuration	A	B	**C**	SELF	**D**	D*	cells
block	0.000	0.133	**1.000**	0.167	**0.200**	1.000	60
needle, pause 4	—	0.087	**0.957**	0.348	**0.174**	—	23
needle, pause 25	—	0.625	**0.958**	0.750	**0.583**	—	24

R3 fails. A constant-velocity model lands essentially every cell. The design requires success(D) − success(C) to clear a positive margin; the difference is negative in all three configurations, by between 0.37 and 0.80.

R2 fails too, and in the competitor's favour. Across the 47 needle cells, **arm D wins zero paired cells that SELF loses**, while SELF wins four that arm D loses—in each configuration separately. Under injected coupling the same comparison ran 146 to 8 in arm D's favour.

4 Why, as far as we can show it: the carrier cannot stop

The relation is made necessary by **intermittency**. A target that rides continuously moves at constant velocity, and then the mode operator suffices by construction; the *pauses* are what neither a single-entity nor a mode model can represent. On CPU the carrier is a scripted point and the intermittency is exact.

A real arm does not stop when told. Measured over 139 goal-pauses in 20 cells, the arm takes a median of **22** steps to read as stopped after its goal stands still, with a ninetieth percentile of 59 and a maximum

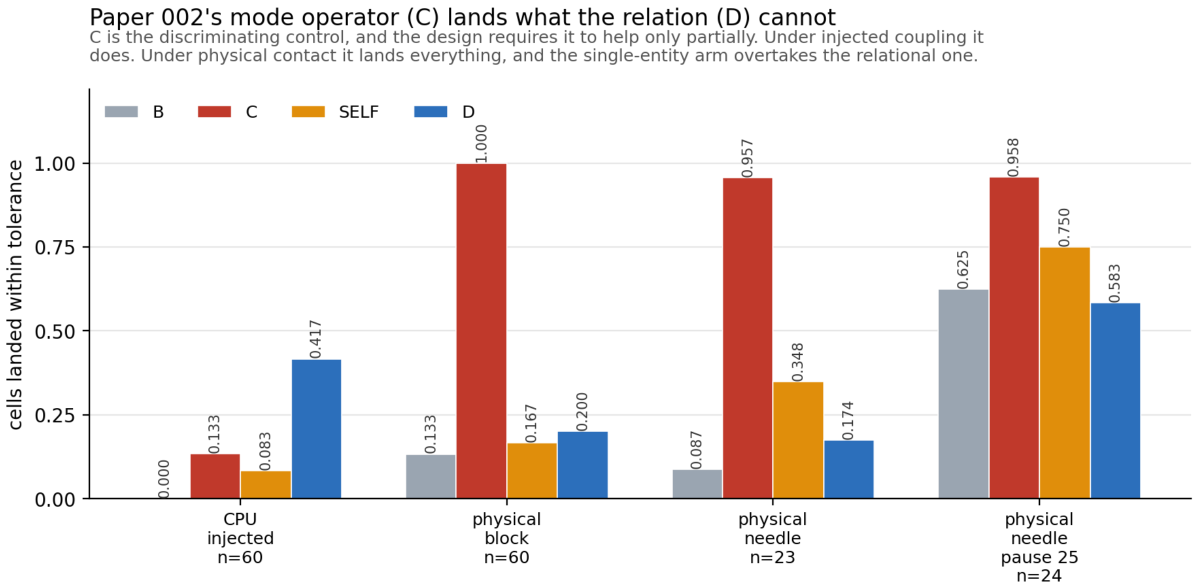

Figure 1: Every arm, injected coupling and three physical configurations. Arm C is the discriminating control, and the design requires it to help only partially. Under injected coupling it does. Under physical contact it lands everything.

of 85—against a commanded pause of 4 steps. **The pause never begins.** What reaches the object is not the commanded square wave but a heavily smoothed version of it, quantified in Section 4.3 as a velocity ripple of 0.174 against the commanded 1.000, and a ride that smooth is precisely the case the mode operator was built for.

4.1 Two repairs, derived in advance, both tried

A better-held object. The manipulator is a needle driver and the block is not what it was built to grip, so the needle is the favourable case: held 68 steps at the median against the block's 52 over the matched cells (54 over all 60), with a tenth percentile of 56 against 16. Arm C: **0.957**. The mode operator still lands everything, and on the needle the single-entity arm overtakes the relational one.

Restoring the pause. The rule was written before the settling time was measured: *the pause must exceed the arm's settling time, plus the gate's minimum ride so that the stillness is observable*—both quantities already declared. That gives **25 steps** at the median. The needle's 68-step carry has room for one such pause where the block's does not. Run at 25: arm C moves from **0.957 to 0.958**.

4.2 The long-pause configuration does not cleanly test what it was built to test

Arm B rises from 0.087 to 0.625. Arm B is a zero-order aim and succeeds when the target barely moves during the dispense. With 25 steps of pause in a 29-step cycle most of the carry is dead time, so most commit windows land where nothing is happening and the trivial arm wins. Every arm rose except C, which had nowhere to rise to.

This was written down as one of three possible readings **before** the run: *"C drops but B is also high → the task setup is broken and eligibility must exclude pause-window commits."* Section 4.4 confirms it on CPU—the same signature reproduces under injected coupling, so this row's anomaly belongs to the commit policy and needs no explanation from the physics.

It is reported as a limitation of that row rather than repaired, because repairing it means changing the preregistered commit policy to exclude dispense windows containing no target motion—a new design, not a fix to this one. **The conclusion therefore rests on the two configurations where arm B is at 0.087 and 0.133**, cells genuinely hard for a zero-order model, and arm C is at 0.957 and 1.000 anyway.

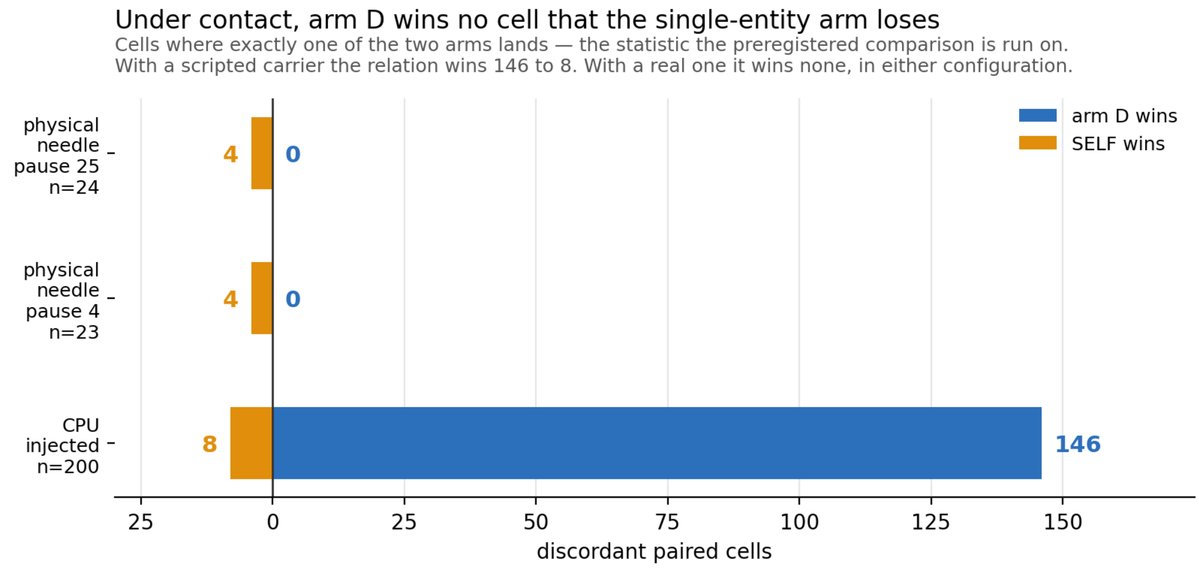

Figure 2: Cells where exactly one of arm D and SELF lands—the statistic the preregistered comparison is run on. With a scripted carrier the relation wins 146 to 8. With a real one it wins none, in either configuration.

4.3 The physical requirement, measured rather than asserted

An earlier draft of this section stated a criterion and derived it from two physical measurements taken separately. That is an inference, not a result, so we tested it. Settling time is a property of the **carrier**, and under injected coupling the carrier is ours to specify: we added a settling parameter that smooths the commanded velocity so the body coasts for exactly that many steps past a stop, swept it over the configuration where the relation is known to be necessary, and fixed four numbered predictions in writing before implementing any of it.

All four predictions failed. What survives is better supported than what was asserted, one claim is withdrawn outright, and one previously open question is closed.

settling	B	C	SELF	D	D : SELF discordant
0—the scripted point	0.000	0.133	0.083	0.417	21 : 1
4—the commanded pause	0.000	0.100	0.017	0.367	21 : 0
9—the commitment latency	0.000	0.350	0.033	0.883	51 : 0
14—the schedule's period	0.000	**0.917**	0.017	0.733	43 : 0
22—the arm's measured settling	0.000	**0.800**	0.000	0.450	27 : 0

The scale is the carrier's duty-cycle period, not its commanded pause. Arm C does not move until settling reaches 9 and reaches 0.917 at **14**, which is the sum of the burst and pause durations. Smoothing does not eat the pause from one end; it low-pass filters the whole waveform, and the intermittency survives as a **ripple** in the carrier's velocity whose size falls as the window approaches the period.

The variable is that ripple, and arm C tracks it at $r = -0.940$.

settling	0	4	6	9	14	22
velocity ripple	1.000	0.800	0.571	0.400	0.067	0.174
arm C	0.133	0.100	0.283	0.350	**0.917**	**0.800**

The ripple is not monotone in settling—a boxcar cancels a periodic signal exactly only when its width is an integer multiple of the period—and arm C is not monotone either, scoring lower at settling 22 than

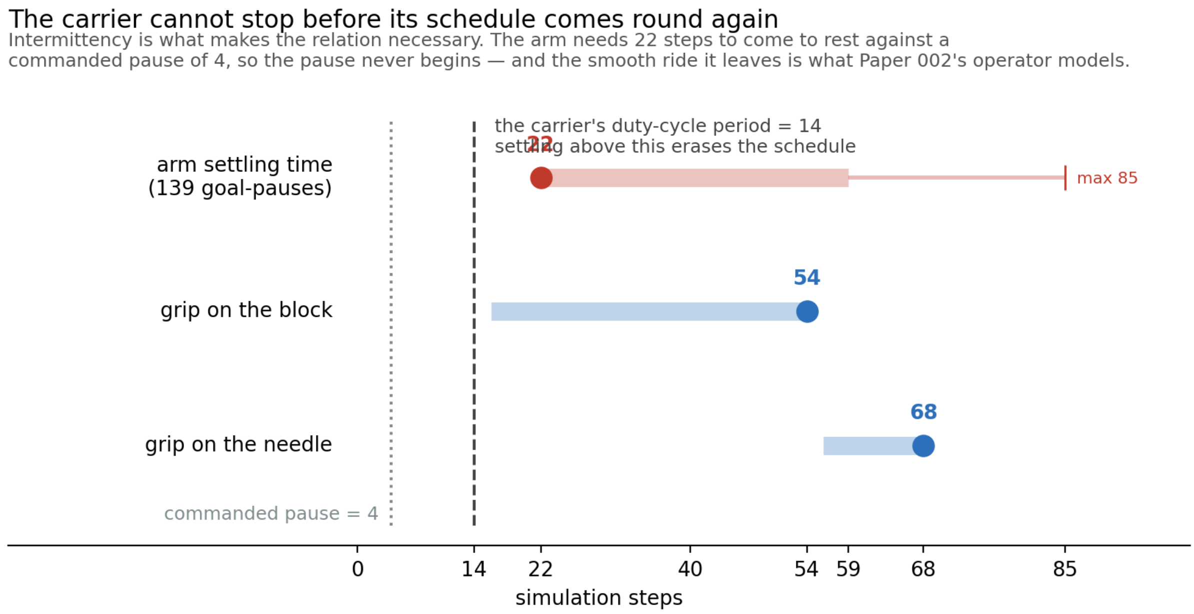

Figure 3: The arm's settling time against its grip duration and the carrier's duty-cycle period. Intervals are the order statistics actually measured.

at 14 for exactly that reason. This was found by a unit test failing on a stronger claim, and it is why we state the criterion on the ripple rather than on a threshold in settling.

The criterion evaluated at matched ripple, which puts CPU and physical configurations on the same axis:

	period	settling	ripple	arm C
CPU, scripted point	14	0	1.000	0.133
CPU	14	22	0.174	0.800
physical, block	14	22	0.174	**1.000**
CPU	35	22	0.435	0.350
physical, needle, long pause	35	22	0.435	**0.958**

The block configuration is accounted for: 1.000 against a predicted 0.800. **The long-pause configuration is not—0.958 against 0.350 at the same ripple**, a gap of about 0.61. We report the model as accounting for one physical configuration and not the other, rather than as the explanation of both.

4.4 What settling time does not explain

Arm D does not collapse under a smoothed carrier. At the arm's own measured settling of 22, on CPU, arm D scores 0.450 against arm B's 0.000, engages on 0.52 of cells, and beats the single-entity arm on **27 discordant pairs to 0**. Physically, at the same settling time, it scored 0.200 and lost that comparison **0 to 4**.

So settling time accounts for arm C's recovery and accounts for **neither** arm D's physical score **nor** the reversal against SELF. We state this rather than leave the simpler account standing. "A real arm cannot pause, therefore the relation dies" was one sentence that appeared to explain everything; it explains arm C, and the rest is open.

One of the two open items closed on a second sweep. Running the same configuration with the derived pause applied reproduces the physical long-pause row's signature on CPU: arm B rises from 0.000 to 0.333, SELF rises to meet arm D, and the discordant count stops being one-sided (27 : 0 becomes 9 : 7, and 7 : 10 at settling 14). Section 4.2's demotion of that row is therefore not a hedge—the anomaly is

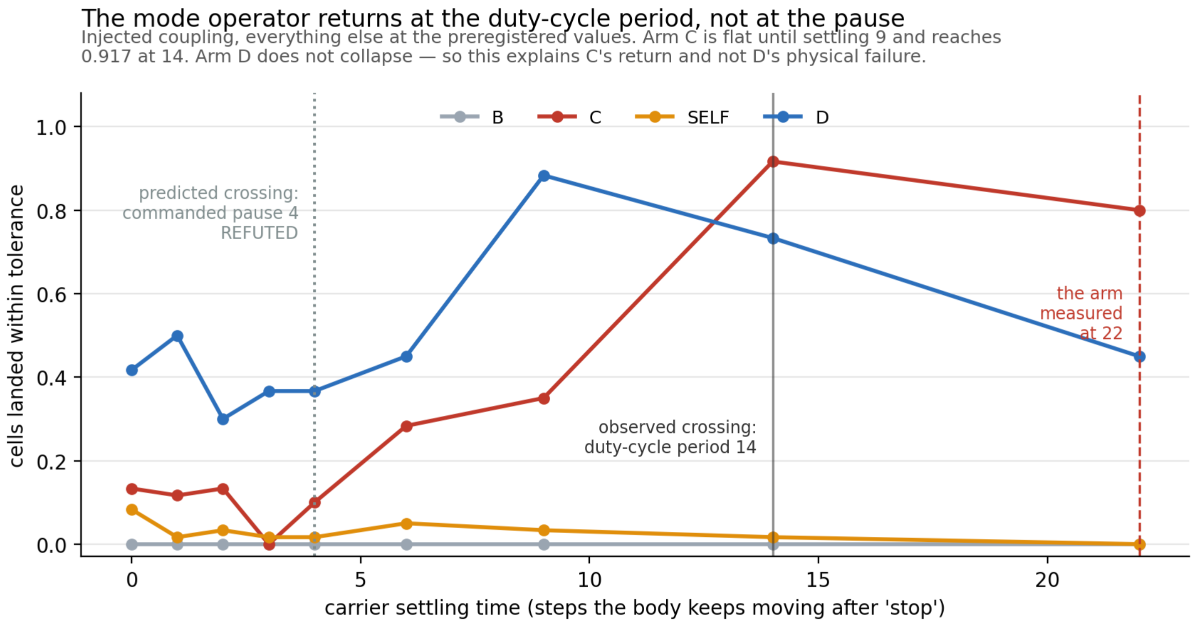

Figure 4: The predicted crossing, at the commanded pause, is marked and labelled refuted. The observed crossing is at the duty-cycle period. Arm D does not collapse.

the commit policy. The predicted threshold for that sweep was $B \geq 0.40$ and the measured value was 0.333, so the prediction is recorded as **failed**; the claim it tested is confirmed by the direction and by three further statistics.

What remains open is narrower and worth naming precisely:

1. **The short-pause reversal.** Physically at a commanded pause of 4, SELF beat arm D 0.348 to 0.174 and won the discordant comparison 4 to 0. On CPU at settling 22 with the same short pause, SELF scores 0.000 and loses 27 to 0.
2. **Arm C where the criterion says it should not be.** At settling 22 against a period of 35 the corrected criterion predicts partial intermittency, and CPU agrees—arm C sits at 0.350. Physically, at the same two numbers, it landed 0.958.

Both point one way: the physical scene makes the target's motion *more* constant-velocity, and the single-entity arm *more* effective, than a smoothed carrier alone accounts for. One nameable and untested hypothesis for its direction: the smoothing models the **arm** tracking its goal, not the **object** tracking the arm. A grasped object is filtered a second time by the compliance of the grip, so the target's velocity in the physical scene should be smoother than this model makes it—which is the direction the discrepancy runs. Testing it means instrumenting the object's velocity against the end effector's, which is GPU work and is not done here.

5 What the negative result is evidence for

The mode/relation boundary sits further out than the taxonomy assumes. The coupling was designed specifically to lie outside the mode operator, and under injected coupling it did—arm C at 0.133 against arm D's 0.417. Under physical contact the same coupling, produced by real dynamics rather than arithmetic, is absorbed completely. What changed is not the relation but the *smoothness of the carrier*, and real actuators are smooth.

This is a reality-gap result of an unusual kind. The literature on the gap is overwhelmingly about *policies* that fail to transfer [6, 12]; Tan et al. [11] find that modelling actuator dynamics is what closes it for locomotion. Here it is a *task property* that fails to transfer, and the direction is inverted: actuator dynamics do not degrade the phenomenon, they delete it. A policy gap is narrowed by more faithful simulation. This one is not, because the more faithful simulation is the one in which the phenomenon is absent—and domain randomisation offers no route, since what is needed is *shorter* settling, not *uncertain* settling.

The program's governing principle returns the answer this paper did not want. Intelligence chooses the smallest sufficient change; in this scene the smallest sufficient change is a mode expansion. A relational expansion here would be a larger change that buys nothing.

The gap between injected coupling and physical contact is total on the decisive comparison, not merely large: 146 discordant pairs to 8 in the relational arm's favour with a scripted carrier, and 0 to 4 against it under contact. A result about arms under injected coupling licenses nothing about arms under contact, and this is the sharpest instance of that we have.

Benchmark-design work diagnoses this class of problem from the outside, after the fact [2, 3, 10]. Here the control fired *inside a preregistered design that named it in advance* as the discriminating control. A design that cannot lose to its own control has not tested anything.

6 Instrumentation, and what went wrong in it

A negative result's credibility rests on whether the instrument would have shown a positive one [5]. Two arguments that it would.

Arm D* lands 1.000 of physical cells. The oracle arm succeeds everywhere arm D fails. The task is solvable from relational information; what fails is estimating it, and the reason it fails is that there is nothing to estimate that the mode operator has not already got.

Arm D does engage. Marginal engagement is 0.27 under contact against 0.23 on CPU, so the gate is not silently refusing to fire. When it acts it is right 0.375 of the time against 0.78 on CPU.

Eleven defects were found and corrected during instrumentation, each recorded where it occurred, and three of them reversed a previously recorded result. The two that matter most for reading this paper:

- An off-by-one in the capture-displacement statistic threw a captured target one body-step past its carrier, so the pauses sorted into the far field and the gate appeared to fire on capture at 1.00. Corrected, the proximity path abstains on *every* capture cell, and capture enters the gate through a second form of positive evidence—carriage—instead.
- Every apparent "ejection" in the first physical sweep, at 36 to 72 mm, was an environment auto-reset being read as an object being flung. Three hypotheses had been built and tested on that artefact before the termination flags were read.

7 What this paper does not claim

- **The relational hypothesis was not rejected at $\alpha = 0.05$.** No confirmatory cell was ever run. The physical runs are calibration, and the preregistration excludes calibration from confirmatory estimates in three separate places. A calibration result does not become confirmatory by turning out to be decisive.
- **The capability-crossing hypothesis was never tested.** Its variant set required a grading variable, reference speed was measured non-functional as one, and no replacement was chosen before the design closed.
- **Gate specificity and non-regression are CPU-only, and structurally so.** Every control condition works by overriding the target's motion, and under contact physics decides it—ten cells requested as static had the arm grasp the block and carry it 35 to 186 mm. They are untested physically, not failed.
- **The injected-coupling result is not overturned.** It was always labelled a comparison of arms under injected coupling with a scripted carrier, and it remains true of scripted carriers.

- **One scene, one manipulator, one relation.** The requirement set generalises as a checklist; the failure does not generalise beyond manipulators whose settling time exceeds their carrier's duty-cycle period.
- **The mechanism is partial.** Section 4.4: the sweep accounts for the mode operator's recovery and not for the relational arm's physical collapse.
- **Observation noise is 0.00 mm per step**, which means this scene has no noise model rather than that noise is small. The gate's margin against noise is unmeasured.

8 What would reopen the cell

Not a larger sample and not a threshold. **A carrier that can come to rest inside the time it holds an object**, so that the intermittency the design needs is physically present at a usable rate. Candidates, in order of cost: a second object driven directly rather than through a manipulator; a conveyor or turntable with a controllable stop; a manipulator with a settling time short relative to its duty-cycle period.

The locked design applies unchanged to any of them. Everything measured here is calibration and should be re-derived rather than inherited.

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